
PRINCIPLES TO FINANCE



*The Finance of the Firm:
An Introduction for First-Year Students*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!

Maths You Will Use

PERCENTAGES

A percentage is a number out of 100. To find 12% of Tk. 500, multiply: $500 \times 0.12 = \text{Tk. } 60$. To turn 12% into a decimal, divide by 100.

PERCENTAGE CHANGE

Percentage change = $(\text{new value} - \text{old value}) / \text{old value} \times 100$. A rise from 40,000 to 48,000 is $8,000 / 40,000 = 0.20$, which is 20%.

RATIOS AND AVERAGES

A ratio compares two numbers by division. An average is the sum of the values divided by how many values there are.

REARRANGING AN EQUATION

Whatever you do to one side, do to the other. From $FV = PV \times 1.1$, divide both sides by 1.1 to get $PV = FV / 1.1$.

POWERS

1.1 to the power 3 means $1.1 \times 1.1 \times 1.1 = 1.331$. A negative power means division: 1.1 to the power -3 is $1 / 1.331 = 0.7513$.

ROUNDING

Keep at least four decimal places in factors, and round money only in the last step. Small differences in the last digit are normal.

READING A GRAPH

The horizontal axis shows one quantity, such as time. The vertical axis shows another, such as money. Each point shows the two values together.

TO THE READER

This book teaches how a business finds, uses and controls its money. It assumes no earlier knowledge of business. If you can do arithmetic and basic algebra, you can follow every step. A word you have not met will be explained when it first appears.

Each page has a wide column of short notes on the left. An italic note repeats the idea of the paragraph beside it in one line. A note in upright type, with a bold word at its start, gives the meaning of a new term. A word in bold in the main text is a new term, set in bold once only. Small raised numbers point to footnotes at the foot of the page. All terms are collected again in the glossary at the back.

Money is written in Bangladeshi taka, shown as Tk. Large sums use the South Asian grouping: Tk. 1,00,000 means one lakh, which is one hundred thousand, and Tk. 10,00,000 means ten lakh, which is one million. Laws, taxes and rates change; where a figure is an assumption it is said to be one. The Bakery in the case files is an invented business, used for illustration. It is not a real firm.

Each chapter begins with a puzzle, builds an idea in steps, and ends with a summary, a problem, common mistakes, a true story from history, and review questions. Short answers are printed at the back. Work each calculation with a pencil before you read the answer.

Further Reading

Brealey, R. and Myers, S. *Principles of Corporate Finance*. A standard text on investment, risk and valuation.

Ross, S., Westerfield, R. and Jordan, B. *Fundamentals of Corporate Finance*. A clear text for beginners.

Van Horne, J. and Wachowicz, J. *Fundamentals of Financial Management*. Strong on working capital and leverage.

Gitman, L. *Principles of Managerial Finance*. Many worked examples.

Jensen, M. and Meckling, W. (1976). Theory of the firm: managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*. The founding paper of agency theory.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*.

Profit maximisation. Aiming for the largest possible profit. (*Chapter I*)

Required rate of return. The yield investors need from a bond of this risk, set by the market. (*Chapter VI*)

Required return. The minimum return an investor needs to accept an investment. (*Chapter IV*)

Retained earnings. Profit that the firm keeps and reinvests instead of paying out as dividends. (*Chapter VIII*)

Return. The gain from an investment, shown as a percentage of the amount invested. (*Chapter IV*)

Risk. The chance that the actual return differs from the expected return. (*Chapter IV*)

Risk premium. The extra return investors require for taking risk instead of a safe investment. (*Chapter IV*)

Security. A paper or record that gives its holder a claim on money from a borrower or a company. (*Chapter VI*)

Share. A unit of ownership in a company; also called common stock. (*Chapter VI*)

Shareholders. The people who own shares, and so own part of a company. (*Chapter I*)

Standard deviation. A measure of how widely returns spread around the expected value. (*Chapter IV*)

Sunk cost. Money already spent that cannot be recovered whatever is decided. (*Chapter III*)

Term loan. A loan from a bank or other lender for a fixed period, repaid in instalments. (*Chapter VIII*)

Time value of money. The idea that a sum of money is worth more now than the same sum later. (*Chapter II*)

Total leverage. The combined effect of operating and financial leverage on profit after interest as sales change. (*Chapter V*)

Trade credit. Goods bought now and paid for later, which is a free or cheap loan from the supplier. (*Chapter VII*)

Wealth maximisation. Aiming to make the owners' investment as valuable as possible. (*Chapter I*)

Working capital. The money a firm needs for daily operations, held in cash, stock and money owed by customers. (*Chapter VII*)

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity. (*Chapter VI*)

A business can earn a profit and still run short of money.

Finance is about finding money and using it well.

Finance. The way a business finds, uses and controls its money.

Accounting looks back; finance looks forward.

Financial manager. The person who decides how a firm finds and uses its money.

Four kinds of decision cover the whole task.

Investment decision. A choice about which assets, such as machines or shops, the firm should buy.

Financing decision. A choice about how to raise money, from owners or from lenders.

Dividend. A share of profit paid out to the owners of a company.

A PUZZLE ABOUT MONEY

A baker sells bread for a month. At the end, her records show a profit of Tk. 20,000. On Friday the flour supplier asks to be paid, and the bank balance is Tk. 4,000. How can a business make a profit and still be unable to pay its bills? The answer lies in the way money moves through a business. The study of that movement is the subject of this book.

WHAT FINANCE MEANS

Finance means the way a business finds money, uses money and controls money. A business needs money to begin, for an oven, flour and rent. It needs more money to continue, for wages and repairs. Finance asks three plain questions. Where will the money come from? Where will it be placed? How will we know it was placed well?

Finance is close to two other subjects. Economics explains how people and markets make choices when resources are limited. Accounting records what has happened in a business. Finance uses both: it takes the records of the past and the ideas of economics to decide what the business should do next.

The person who makes these choices is the **financial manager**. In a small firm the owner does this work. In a large firm a team does it. The financial manager is also a manager in the ordinary sense, so finance and management cannot be separated: every plan of the firm, whether to hire, to expand or to buy, needs money, and the financial manager says whether the money can be found and what it will cost.¹

Before moving on. Finance means finding, using and controlling money. It looks forward, where accounting looks back. The financial manager makes the choices about money.

THE DECISIONS OF THE FINANCIAL MANAGER

The first decision concerns what to buy. An **investment decision** means a choice about which assets the firm should buy, such as a machine, a shop or a delivery van. An asset is anything of value that the firm owns. The second decision concerns where the money comes from. A **financing decision** means a choice about how to raise money: from the owners, from lenders, or from the profits the firm has kept.

The third decision concerns daily money. The working-capital decision means a choice about how much cash, stock and credit the firm needs from day to day to keep running. The fourth is the dividend decision. A **dividend** is a share of profit paid to the owners. The firm must decide how much profit to pay out and how much to keep for growth.

THE FOUR DECISIONS

Decision	The question it answers	Example
Investment	What should we buy?	A second oven
Financing	Where does the money come from?	A bank loan or owners' savings
Working capital	How do we keep daily money flowing?	How much flour to hold in stock
Dividend	How much profit do we pay out?	Pay owners now or keep the money for growth

Glossary

Agency costs. The costs of watching agents and of the losses when they act against the owners. *(Chapter I)*

Agency problem. The risk that agents act for themselves instead of for the owners. *(Chapter I)*

Agent. A person who acts on behalf of another. *(Chapter I)*

Annuity. A series of equal payments made at equal intervals. *(Chapter II)*

Annuity due. An annuity whose payments are made at the start of each period. *(Chapter II)*

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold. *(Chapter VI)*

Bond status. Whether a bond sells below, at or above its face value. *(Chapter VI)*

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs. *(Chapter IV)*

Capital budgeting. Planning and judging large spending whose benefits last for several years. *(Chapter III)*

Capital market. The market where long-term funds are raised by selling shares and bonds. *(Chapter VIII)*

Cash conversion cycle. The days between paying for materials and receiving cash from customers. *(Chapter VII)*

Cash discount. A reduction in price offered for payment within a short time. *(Chapter VII)*

Cash flow. Money actually received or paid by the firm. *(Chapter III)*

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return. *(Chapter IV)*

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid. *(Chapter VIII)*

Compound interest. Interest calculated on the original sum plus the interest already earned. *(Chapter II)*

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money. *(Chapter VIII)*

Coupon rate. The yearly interest rate on the face value, set when the bond is issued. *(Chapter VI)*

Debenture. A long-term bond issued by a company, usually paying fixed interest. *(Chapter VIII)*

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT. *(Chapter V)*

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales. *(Chapter V)*

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales. *(Chapter V)*

Discount rate. The yearly rate used to reduce a future sum to its present value. *(Chapter II)*

Dividend. A share of profit paid out to the owners of a company. *(Chapter I)*

Efficient market. A market in which prices already reflect the available information. *(Chapter IV)*

Expected value. The average of possible returns, each weighted by its probability. *(Chapter IV)*

they act against the owners.

between lenders and owners, because owners may take risks with money that lenders supplied.



WORKED CASE: THE BAKERY

WORKED EXAMPLE: WHY DID CASH FALL WHEN THE BAKERY MADE A PROFIT?

SITUATION. *Case file, illustrative.* In January 2025 the owners put Tk. 5,00,000 into the Bakery. Oven and fittings cost Tk. 3,50,000, leaving cash of Tk. 1,50,000. In its first month the Bakery sold goods worth Tk. 1,50,000, of which Tk. 60,000 was sold to a café on 30 days' credit. Costs of Tk. 1,30,000 were paid in cash.

QUESTION. What was the profit, and what happened to cash?

REASONING. Profit is sales minus costs: 1,50,000 - 1,30,000 = Tk. 20,000. Cash received was 1,50,000 - 60,000 = Tk. 90,000. Cash paid was Tk. 1,30,000. Cash change is 90,000 - 1,30,000 = minus Tk. 40,000.

CONCLUSION. The Bakery made a profit of Tk. 20,000, yet its cash fell by Tk. 40,000, from 1,50,000 to Tk. 1,10,000. The Tk. 60,000 owed by the café is profit that has not yet become cash.

CHECK. Profit 20,000 less the unpaid 60,000 gives minus 40,000, which matches the cash change.

CLOSING THE CHAPTER

SUMMARY

Finance is the way a firm finds, uses and controls money. The financial manager makes investment, financing, working-capital and dividend decisions. The sound goal is wealth maximisation, because it counts time and risk. Where owners and managers are different people, agency costs arise and must be controlled.

THINK IT THROUGH

A manager can raise this year's profit by Tk. 50,000 by stopping all repairs to the ovens. Explain why this may reduce the owners' wealth.

WHERE PEOPLE GO WRONG

Many students think profit and cash are the same; the case above shows they are not. Others think wealth maximisation means ignoring profit; it does not, since profit is one source of future cash.

IN HISTORY

In 1602 the Dutch East India Company offered shares to the public, and ordinary people became part-owners of a business run by others far away. In 1776 Adam Smith wrote that directors of such companies, handling other people's money, could hardly be expected to watch it as anxiously as owners watch their own.²

REVIEW QUESTIONS

1. What is meant by finance?
2. Name the four decisions of a financial manager and say what each answers.
3. Why is profit maximisation not the ultimate goal of a firm?

¹ Management is the wider task of organising people and work to reach a goal. Finance is the part of that task that deals with money.

Profit and cash are different things.

Answers to Review Questions

CHAPTER I

1. Finance is the way a business finds, uses and controls money; it asks where money comes from, where it is placed and how its use is judged.
2. Investment (what to buy), financing (where money comes from), working capital (daily money), dividend (how much profit to pay out).
3. It ignores timing, risk and the difference between profit and cash, and may harm the future; wealth maximisation counts all three.

CHAPTER II

1. A sum today is worth more than the same sum later, because money can earn, prices rise and the future is uncertain.
2. A series of equal payments at equal intervals; ordinary (end of period), due (start of period), and perpetuity (never ends).
3. $P = \text{Tk. } 12,434.26$; $Q = \text{Tk. } 10,383.17$. P is worth more today.
4. $1,00,000 \times 5.8666 = \text{Tk. } 5,86,660$.

CHAPTER III

1. Planning and judging large, long-lived spending; necessary because sums are large, benefits long, decisions hard to reverse and the future of the firm depends on them.
2. Generate proposals; estimate cash flows; evaluate; select; implement; review.
3. $\text{Payback} = 20,000 / 8,000 = 2.5$ years. $\text{NPV} = 8,000 \times 2.48685 - 20,000 = \text{Tk. } -105.18$. It is negative, so reject.

CHAPTER IV

1. The chance that the actual return differs from the expected return.
2. Business risk comes from operations and exists without debt; financial risk is the extra risk from borrowing.
3. $\text{CV of } M = 4/10 = 0.40$; $\text{CV of } N = 6/20 = 0.30$. N is less risky for each unit of return.
4. $(54 - 50 + 2) / 50 = 12\%$.

CHAPTER V

1. Operating: fixed operating costs magnify the effect of sales on EBIT. Financial: fixed financing costs magnify the effect of EBIT on EBT. Total: the two combined, from sales to EBT.
2. It magnifies both profit and loss, so it raises risk; managers must balance the gain against the danger of failing to meet fixed costs.
3. $\text{Contribution} = 2,000 \times 20 = 40,000$; $\text{EBIT} = 20,000$; $\text{EBT} = 15,000$. $\text{DOL} = 2.00$; $\text{DFL} = 1.33$; $\text{DTL} = 2.67$.

CHAPTER VI

1. A long-term loan split into tradable pieces; face value, coupon rate and maturity are its key features.
2. Coupon below required return: discount. Equal: par. Above: premium.

A PUZZLE ABOUT WAITING

A friend offers you a choice: Tk. 1,000 today, or Tk. 1,000 in one year. Most people take the money today. Now the offer changes: Tk. 1,000 today, or Tk. 1,100 in one year. The choice is harder. Somewhere between 1,000 and 1,100 lies a point where you are indifferent. That point is the key to this chapter.

THE TIME VALUE OF MONEY

The **time value of money** means that a sum of money is worth more now than the same sum later. There are three reasons. Money held today can be placed in a bank or a business and earn a return. Prices usually rise, so later money buys less. And a promise of future money may not be kept, so waiting carries risk.

Suppose you place Tk. 1,000 in a bank that pays 10% a year. **Interest** means the payment for the use of money. After one year you hold $1,000 + 100 = \text{Tk. } 1,100$. In the second year the bank pays 10% on 1,100, not on 1,000. You hold $1,100 + 110 = \text{Tk. } 1,210$. Interest earned on earlier interest is called **compound interest**.

You should now be able to say: *compound interest grows a sum faster than simple interest, because each year's interest earns interest too.*

MOVING A SUM FORWARD AND BACK

The **future value** (FV) is what a sum today will be worth later. The **present value** (PV) is what a future sum is worth today. The **discount rate** is the rate used to move a future sum back to today. We write r for the rate as a decimal and n for the number of years.

Rule: $\text{future value} = \text{present value} \times (1 + r)^n$. $\text{Present value} = \text{future value} / (1 + r)^n$. The two are opposites.

WORKED EXAMPLE: SINGLE SUM

GIVEN. $\text{PV} = \text{Tk. } 1,000$; $r = 10\%$ a year; $n = 2$ years.

FIND. the future value, and then the present value of that future value.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$; $\text{PV} = \text{FV} / (1 + r)^n$.

STEPS. $\text{FV} = 1,000 \times 1.10^2 = 1,000 \times 1.21 = \text{Tk. } 1,210$. $\text{PV} = 1,210 / 1.21 = \text{Tk. } 1,000$.

ANSWER. Tk. 1,000 today equals Tk. 1,210 in two years at 10%.

CHECK. This matches the year-by-year result above: 1,100, then 1,210. Moving forward and back returns the starting sum.

WORKED EXAMPLE: A MESSIER CASE: Tk. 25,000 FOR 5 YEARS AT 12%

GIVEN. $\text{PV} = \text{Tk. } 25,000$; $r = 12\%$; $n = 5$.

FIND. FV.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$.

STEPS. $1.12^5 = 1.7623$. $\text{FV} = 25,000 \times 1.7623 = \text{Tk. } 44,059$.

ANSWER. Tk. 25,000 grows to about Tk. 44,059 in five years.

CHECK. Simple interest would give $25,000 + 5 \times 3,000 = \text{Tk. } 40,000$. Compound interest must be larger, and it is.

Would you take Tk. 1,000 today or Tk. 1,000 in a year?

Money today can earn; money later cannot.

Time value of money. The idea that a sum of money is worth more now than the same sum later.

Go slowly here.

Interest. The payment a borrower makes for the use of money, shown as a percentage of the sum.

Compound interest. Interest calculated on the original sum plus the interest already earned.

Future value pushes money forward; present value pulls it back.

Future value. What a sum of money today will be worth at a later date, after earning interest.

Present value. What a sum due in the future is worth today.

Discount rate. The yearly rate used to reduce a future sum to its present value.

Common slips with single sums.

Common slips: using 10 instead of 0.10 in the formula; forgetting that n counts years, so 6 months is $n = 0.5$; and dividing when the question asks you to multiply.

STEPS. Lease: $30,000 \times 0.70 = 21,000$ each year. $PV = 21,000 \times 3.38721 = 71,131$. Purchase outflows: Year 1: $31,500 - 0.30 \times 35,000 = 21,000$; Year 2: $31,500 - 0.30 \times 32,500 = 21,750$; Year 3: $31,500 - 0.30 \times 30,000 = 22,500$; Year 4: $31,500 - 0.30 \times 27,500 = 23,250$. $PV = 74,727$.

ANSWER. Leasing has the lower present value, Tk. 71,131 against Tk. 74,727 for buying, so lease the van.

CHECK. Each yearly lease outflow (21,000) is below or equal to each purchase outflow, so leasing must have the lower present value. This test is not always possible; here it is.

Exceptions: this comparison leaves out the value of the van at the end of four years, which a buyer keeps. If that value is large, buying may win. The residual value must be estimated and added. Maintenance, insurance and risk of the asset becoming old also matter.

CLOSING THE CHAPTER

SUMMARY

A lease lets the lessee use an asset owned by the lessor. Operating leases are short and cancellable; financial leases are long and work like loans. The lease-or-buy decision compares the present value of after-tax outflows at the after-tax cost of debt.

THINK IT THROUGH

Explain why a firm needing a computer for six months would prefer an operating lease to a financial lease.

WHERE PEOPLE GO WRONG

Students discount at the before-tax rate, or forget the tax saving on depreciation and interest. Others ignore the value of the asset at the end.

IN HISTORY

In 1952 a firm in San Francisco, the United States Leasing Corporation, began to lease equipment to businesses that wanted to use machines without buying them. Leasing then spread as a standard way of financing equipment.

REVIEW QUESTIONS

1. What is leasing? Distinguish operating from financial leases.
2. State the four steps of the lease-or-buy decision.
3. Why is the after-tax cost of debt used as the discount rate?

CLOSING THE CHAPTER

SUMMARY

Money has a time value because it can earn, prices rise and promises carry risk. Future value moves money forward and present value moves it back. Annuities use short formulas; mixed streams are moved one payment at a time.

THINK IT THROUGH

Stream P pays Tk. 5,000 at the end of each of 3 years. Stream Q pays 2,000, 4,000 and 7,000 in those years. Find the present value of each at 10% and say which is worth more today.

WHERE PEOPLE GO WRONG

Students add payments from different years as if they were equal. Others use an ordinary annuity formula for payments made at the start of the period.

IN HISTORY

In 1202 the Italian mathematician Leonardo of Pisa, called Fibonacci, published *Liber Abaci*, which taught merchants of Europe the Hindu-Arabic numerals. It included problems on interest and on the value of money at different times, tools for the traders of his day.

REVIEW QUESTIONS

1. What is meant by the time value of money?
2. What is an annuity, and what are its kinds?
3. Find the present values of streams P and Q at 10%. Which is better?
4. Tk. 1,00,000 is saved at the end of each year for 5 years at 8%. What is the future value?

A PUZZLE ABOUT TWO MACHINES

Machine A and Machine B each cost Tk. 10,000. A returns Tk. 5,000 a year for three years. B returns Tk. 2,000, then 4,000, then 10,000. Both return Tk. 15,000 and Tk. 16,000 in total. Which should the owner buy? A simple count of total money received favours B, but A returns money sooner. We need a method that treats both the size and the timing of money fairly.

WHAT CAPITAL BUDGETING IS

Capital budgeting means planning and judging large spending whose benefits last for several years, such as a new oven, a shop or a factory. It is necessary for four reasons. The sums are large. The benefits last a long time. Such decisions are hard to reverse, since a machine cannot be returned once installed. And they shape what the firm can do in future, so a poor choice weakens the firm for years.

THE STEPS OF THE PROCESS

First, the firm generates proposals, with ideas coming from managers, workers and customers. Second, it estimates the cash flows each proposal will cause. Third, it evaluates each proposal using a technique. Fourth, it selects and approves the best. Fifth, it carries out the project. Sixth, it reviews the results later and compares them with the estimates, so that future estimates improve.

CASH FLOWS THAT COUNT

A **cash flow** is money actually received or paid. The project's cash flows come in three groups: the initial outlay, which is the cost of starting; the operating cash inflows, which are the extra yearly receipts after tax; and the terminal cash flow, such as the sale of the machine at the end.

Only **incremental** cash flows count, which means the change caused by the project. A **sunk cost** is money already spent, such as a survey paid for last year; it is ignored, since it cannot be recovered. A use of an owned resource counts at what it could earn elsewhere. Depreciation is not a cash payment, so it is not counted as a cash flow, though it affects the tax paid.

TWO TECHNIQUES

The **payback period** (BBP) is the time a project takes to return its initial cost. The rule is to accept a project if its payback is shorter than a limit the firm has set, and to prefer the shorter payback. Its strength is simplicity and its attention to how soon the money returns. Its weakness is serious: it ignores the time value of money and it ignores all cash after the payback date.

The **net present value** (NPV) is the present value of the cash inflows minus the initial cost. The rule is to accept a project if NPV is greater than zero, and to prefer the project with the highest NPV. A positive NPV means the project earns more than the discount rate, and so adds to the owners' wealth. This links back to Chapter I: NPV is wealth maximisation made measurable.

Other techniques exist. The internal rate of return finds the discount rate at which NPV is zero. The profitability index divides the present value of inflows by the cost. They are met in later courses.

Two machines cost the same but pay back differently.

Large, long-lasting spending needs careful judgement.

Capital budgeting. Planning and judging large spending whose benefits last for several years.

Capital budgeting follows six steps in order.

Only extra cash caused by the project counts.

Cash flow. Money actually received or paid by the firm.

Incremental. Extra; the change in cash caused by choosing the project.

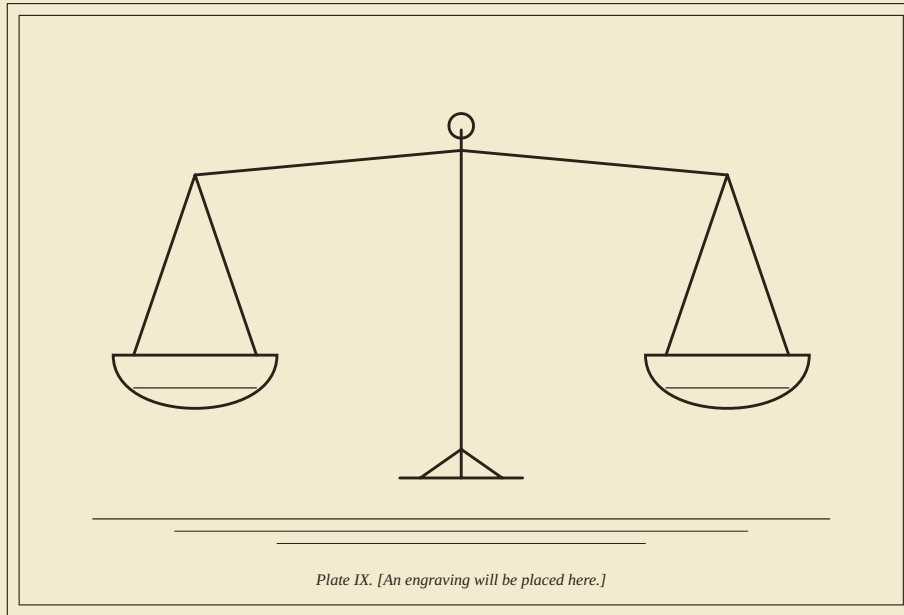
Sunk cost. Money already spent that cannot be recovered whatever is decided.

Payback asks how soon the money returns.

Payback period. The time a project takes to return its initial cost in cash.

NPV measures the wealth a project adds.

Net present value. The present value of a project's cash inflows minus its initial cost.



CHAPTER IX

Leasing

To use a thing is not always to own it.



This chapter explains leasing, the two main kinds of lease, and the steps for deciding whether a firm should lease or buy an asset.

Source	Paid	Vote	Cost to firm	If payment missed
Retained earnings		Existing owners	Same as common stock	

Read the table across: each row describes one source on the six features above.

RAISING MONEY IN THE CAPITAL MARKET

The **capital market** is the market where long-term funds are raised and traded. A company can sell shares to the public for the first time in an **initial public offering** (IPO). It can offer new shares to its present owners in a rights issue. It can sell shares to a few large investors in a private placement. It can issue bonds or debentures. Each method needs approval from the regulator, a document describing the firm, and fees to the people who arrange the sale.

In Bangladesh, shares are traded on the Dhaka Stock Exchange and the Chit-tagong Stock Exchange, and the Bangladesh Securities and Exchange Commission regulates the market. Long-term finance is also supplied by commercial banks, by development finance institutions such as the Bangladesh Development Bank Limited and the Investment Corporation of Bangladesh, and by lenders for infrastructure such as IDCOL. Names and rules change, so check the current position before relying on any of this.

The **cost of capital** is the average return a firm must earn to satisfy all who supplied its long-term money. A firm should accept an investment only if it earns more than this cost. When we judge a project, we assume the project does not change the firm's business risk or financial risk; otherwise the cost of capital would change too.

CLOSING THE CHAPTER

SUMMARY

Long-term sources are debt, preferred stock, common stock and retained earnings. They differ in claim, control, cost and risk. Firms raise them in the capital market through shares, bonds and loans. The cost of capital is the average price of all of them.

THINK IT THROUGH

A firm can borrow at 15% with a tax rate of 30% or raise equity at 18%. Which is cheaper after tax, and why might the firm still not borrow without limit?

WHERE PEOPLE GO WRONG

Students say retained earnings cost nothing. Others forget that interest lowers tax and so compare before-tax debt with equity.

IN HISTORY

In 1694 the English government needed money for war with France. A group of subscribers lent it 1,200,000 pounds and in return was set up as the Bank of England. It was an early case of the public being invited to supply long-term money to a borrower.

REVIEW QUESTIONS

1. Why is long-term financing important for a new business?
2. Name the main sources of long-term money.
3. Solve the Think it through problem.

rowing. Tolls repaid the debt early, and the canal lowered the cost of moving goods between the Atlantic coast and the Great Lakes.

REVIEW QUESTIONS

1. What is capital budgeting, and why is it necessary?
2. List the steps of capital budgeting.
3. Solve the Think it through problem.

Large firms sell securities to many investors.

Capital market. The market where long-term funds are raised by selling shares and bonds.

Initial public offering. The first sale of a company's shares to the public.

Cost of capital is the average price of the firm's money.

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money.

A PUZZLE ABOUT TWO ROUTES

You travel to the city by two routes. Route X takes 40, 45 or 50 minutes. Route Y takes 20, 45 or 70 minutes. Both average 45 minutes. Yet if you must reach an examination on time, you choose X. Y is riskier because its results are more spread out. Finance measures this spread in the same way.

RETURN AND RISK

Return means the gain from an investment as a percentage of the amount invested. Rate of return = (selling price - buying price + cash received) / buying price. If you buy a share for Tk. 100, receive a dividend of Tk. 4 and sell for Tk. 108, the return is $(108 - 100 + 4) / 100 = 12\%$.

Risk means the chance that the actual return will differ from the one expected. Both bad and good surprises are included in the idea, though people worry about the bad.

KINDS OF RISK

Business risk means the risk that comes from the firm's operations: sales may fall, costs may rise, a rival may arrive. It exists even if the firm has no debt. **Financial risk** means the extra risk that comes from borrowing. Interest must be paid whether sales are good or poor, so debt makes the owners' return less certain. Chapter V studies this link.

Risks can also be sorted by whether they can be removed. Risk that belongs to one firm, such as a strike or a failed product, can be reduced by holding shares in many firms. Risk that affects all firms, such as a recession or a rise in interest rates, cannot. Investors are paid mainly for bearing the second kind.

MEASURING RISK

The measure uses four steps. A **probability distribution** lists the possible returns and the chance of each. The **expected value** is the average, with each return weighted by its chance. The **standard deviation** measures the spread: take each return's distance from the expected value, square it, weight it by its chance, add, and take the square root. The **coefficient of variation** divides the standard deviation by the expected value, so it shows risk for each unit of return and allows fair comparison.

WORKED EXAMPLE: INVESTMENTS X AND Y

GIVEN. three possible conditions with probabilities 0.25, 0.50, 0.25. Returns of X: 8%, 12%, 16%. Returns of Y: 4%, 12%, 20%.

FIND. expected value, standard deviation and coefficient of variation of each.

FORMULA. Expected value = sum of (probability x return). Standard deviation = square root of sum of [probability x (return - expected value)²]. CV = standard deviation / expected value.

STEPS. X: expected value = $0.25 \times 8 + 0.50 \times 12 + 0.25 \times 16 = 2 + 6 + 4 = 12\%$. Squared distances: $(8-12)^2 = 16$, 0, 16. Weighted: $0.25 \times 16 + 0 + 0.25 \times 16 = 8$. Standard deviation = square root of 8 = 2.83%. Y: expected value = $1 + 6 + 5 = 12\%$. Squared distances: 64, 0, 64. Weighted sum = 32. Standard deviation = 5.66%.

ANSWER. Both average 12%. X has standard deviation 2.83% and CV 0.24. Y has standard deviation 5.66% and CV 0.47. Y is twice as risky for the same return.

Same average, different spread.

Return is the gain; risk is doubt about the gain.

Return. The gain from an investment, shown as a percentage of the amount invested.

Risk. The chance that the actual return differs from the expected return.

Business risk comes from operations; financial risk from borrowing.

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs.

Financial risk. The extra risk that comes from using borrowed money.

Risk is measured by how far returns spread from their average.

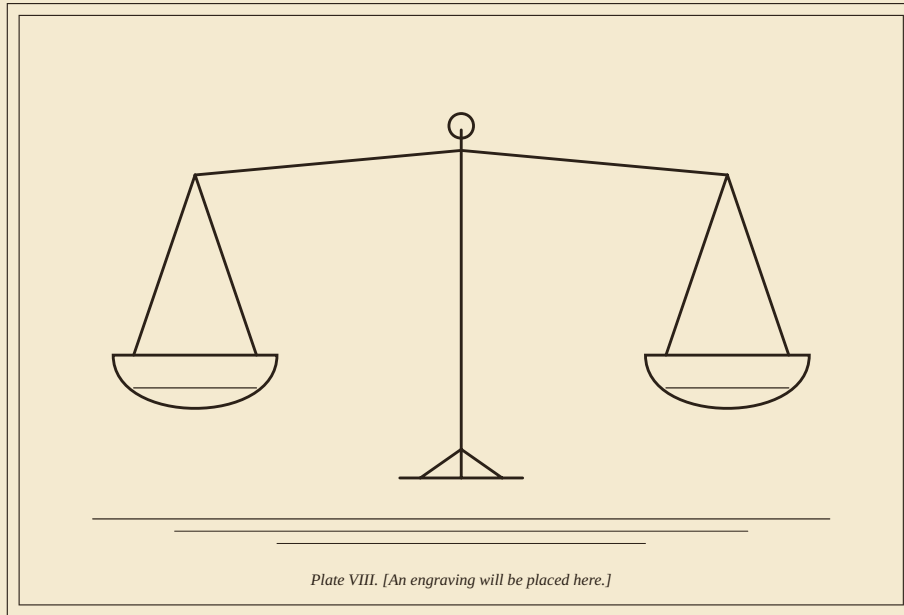
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Probability distribution. A list of possible returns, each with the chance that it occurs.

Expected value. The average of possible returns, each weighted by its probability.

Standard deviation. A measure of how widely returns spread around the expected value.

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return.



CHAPTER VIII

Long-Term Funds

A firm that plans to last must find money that lasts.



This chapter explains the long-term sources of money open to a firm: term loans, bonds, preferred shares, ordinary shares and retained profits. It shows how funds are raised in the capital market and which institutions supply them in Bangladesh.

ANSWER. Not taking the discount costs about 36.7% a year, more than the bank's 14%. Take the discount, and borrow from the bank if cash is needed.
CHECK. For 20 days the cost is $2 / 98 = 2.04\%$; there are 18 such periods in a year, and 2.04×18 is about 36.7.

With terms 1/10 net 40 the cost is $(1 / 99) \times (360 / 30) = 12.12\%$, below 14%, so the firm should pay on day 40. Exceptions: the formula uses simple interest; with compounding the true cost is higher. Paying late costs less in the short run but may harm the firm's name with suppliers.

FLOAT AND THE CASH CYCLE

Float means the time between sending a payment and the receiver having usable funds. It has three parts: mail time, while the cheque travels; processing time, while the receiver records and deposits it; and clearing time, while the banks move the money. A firm wants short float on money it receives and, within honest limits, longer float on money it pays.

The **cash conversion cycle** shows how long money is tied up: inventory days + receivables days - payables days. If the Bakery holds stock for 40 days, waits 30 days for customers and pays suppliers after 20 days, the cycle is $40 + 30 - 20 = 50$ days.

WORKED EXAMPLE: ESTIMATING WORKING CAPITAL

GIVEN. yearly cost of sales Tk. 36,00,000; cash conversion cycle 50 days; 360 days in a year.

FIND. the working capital needed.

FORMULA. Working capital = daily cost $\times$ length of cycle in days.

STEPS. Daily cost = $36,00,000 / 360 = \text{Tk. } 10,000$. Working capital = $10,000 \times 50 = \text{Tk. } 5,00,000$.

ANSWER. The Bakery needs about Tk. 5,00,000 to run its daily operations.

CHECK. A cycle of 50 days is about 50/360 of a year, and 50/360 of 36,00,000 is Tk. 5,00,000.

CLOSING THE CHAPTER

SUMMARY

Working capital keeps daily operations running. Financing is short, medium or long term, and should match the life of the asset. Trade credit has a cost that can be calculated. Float delays money, and the cash conversion cycle shows how much working capital is needed.

THINK IT THROUGH

Terms are 3/10 net 45 and the bank charges 18%. Find the cost of trade credit and decide whether to take the discount.

WHERE PEOPLE GO WRONG

Students forget to divide the discount by 100 minus the discount. Others use the full credit period instead of the days after the discount period.

IN HISTORY

Francesco Datini, a merchant of Prato in Italy who died in 1410, left behind account books and a very large number of letters. They show how medieval traders used credit and bills of exchange to trade without carrying coins, an early lesson in managing the gap between paying and being paid.

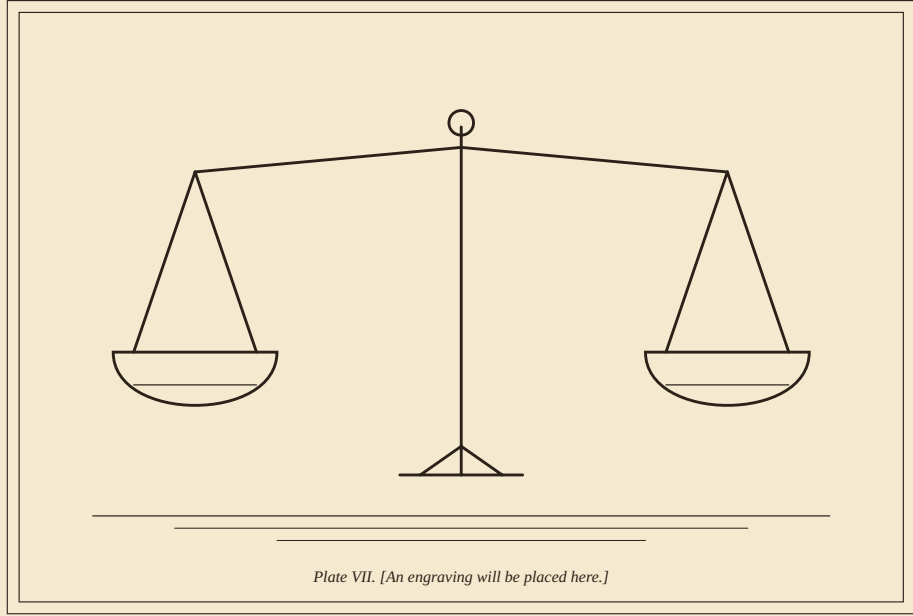
REVIEW QUESTIONS

1. What is risk?
2. Distinguish financial risk from business risk.
3. Solve the Think it through problem.
4. A share costs Tk. 50, pays Tk. 2 and is sold for Tk. 54. Find the return.

Float is the delay between paying and the money moving.

Float. The time between a payment being sent and the money becoming usable by the receiver.

Cash conversion cycle. The days between paying for materials and receiving cash from customers.



CHAPTER VII

Short-Term Money

A firm can fail with a full order book and an empty till.



This chapter explains working capital, the length of financing, the matching principle, the cost of trade credit, the delay called float, and how to estimate the working capital a firm needs.

The same lever that lifts also drops.

Leverage means fixed costs magnify results.

Leverage. The use of fixed costs to magnify the effect of a change in sales on profit.

Fixed cost. A cost that stays the same whatever the level of sales.

Operating leverage. The effect of fixed operating costs on how profit before interest and tax changes with sales.

Financial leverage. The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT.

Total leverage. The combined effect of operating and financial leverage on profit after interest as sales change.

The degree of leverage is a multiplier.

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales.

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT.

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales.

Break-even is where profit before interest and tax is zero.

Leverage raises both reward and risk.

A PUZZLE ABOUT A LEVER

A shopkeeper rents a shop for a fixed Tk. 40,000 a month and pays little for goods. Another buys goods at a high price but pays a low rent. In a month when sales rise by 10%, which shopkeeper's profit rises more? The first, since the rent does not rise with sales. In a month when sales fall by 10%, the same shopkeeper's profit falls more. A fixed cost works like a lever: it magnifies the effect of sales on profit, upward and downward.

WHAT LEVERAGE IS

Leverage means the use of fixed costs, which are costs that do not change when sales change, to magnify the effect of a change in one quantity on another. A **fixed cost** stays the same whatever the level of sales, as rent does. A variable cost rises and falls with sales, as flour does. The word comes from the lever of Archimedes.

Two kinds of fixed cost give two kinds of leverage. **Operating leverage** comes from fixed operating costs, such as rent and machines, and shows how profit before interest and tax (EBIT) responds to sales. **Financial leverage** comes from fixed financing costs, mainly interest, and shows how profit after interest (EBT) responds to EBIT. **Total leverage** combines both.

MEASURING LEVERAGE

The **degree of operating leverage** (DOL) is the percentage change in EBIT for each 1% change in sales. At a given sales level, $DOL = \text{contribution} / EBIT$, where $\text{contribution} = \text{sales} - \text{variable costs}$. The **degree of financial leverage** (DFL) = $EBIT / EBT$. The **degree of total leverage** (DTL) = $\text{contribution} / EBT$, which equals $DOL \times DFL$.

WORKED EXAMPLE: LEVERAGE OF A SMALL FIRM

GIVEN. price Tk. 50 per unit; variable cost Tk. 30 per unit; fixed operating cost Tk. 40,000; interest Tk. 10,000; sales 4,000 units.

FIND. DOL, DFL and DTL, and the effect of a 10% rise in sales.

FORMULA. $DOL = \text{contribution} / EBIT$; $DFL = EBIT / EBT$; $DTL = DOL \times DFL$.

STEPS. Contribution per unit = $50 - 30 = 20$. Contribution = $4,000 \times 20 = 80,000$. $EBIT = 80,000 - 40,000 = 40,000$. $EBT = 40,000 - 10,000 = 30,000$. $DOL = 80,000 / 40,000 = 2.00$. $DFL = 40,000 / 30,000 = 1.33$. $DTL = 80,000 / 30,000 = 2.67$, which equals 2.00×1.33 .

ANSWER. $DOL = 2.00$, $DFL = 1.33$, $DTL = 2.67$. A 10% rise in sales should raise EBT by about 26.7%.

CHECK. At 4,400 units, contribution is 88,000, EBIT is 48,000 (up 20%), EBT is 38,000 (up 26.7%). It matches.

The break-even quantity is fixed operating cost divided by contribution per unit: $40,000 / 20 = 2,000$ units. Below it, the firm makes a loss before interest.

WHY LEVERAGE MATTERS

Leverage raises the reward of good times and the damage of bad times. A firm with high operating leverage, such as one with a costly factory, is hurt badly when sales fall. A firm with high financial leverage must pay interest even in a poor year, so it may fail to pay and lose its assets. From the owner's side, debt

WORKED EXAMPLE: A BOND PAYING INTEREST TWICE A YEAR

GIVEN. face value Tk. 1,000; coupon 6.5% a year paid half-yearly; 2 years; required return 8% a year.

FIND. the value.

FORMULA. Use half-yearly periods: halve the coupon and the rate, and double the years.

STEPS. Interest each half-year = $1,000 \times 6.5\% / 2 = \text{Tk. } 32.50$. Periods = 4. Rate per period = 4%. Annuity factor = 3.62990; single-sum factor = 0.85480. Value = $32.50 \times 3.62990 + 1,000 \times 0.85480 = 117.97 + 854.80$.

ANSWER. The value is Tk. 972.78.

CHECK. The coupon (6.5%) is below the required return (8%), so the value is below 1,000. It is.

SHARES

A **share** means a unit of ownership in a company, also called common stock. The holder receives dividends if the firm pays them and gains if the share price rises. Unlike a bond, a share has no maturity and no promised payment.

BOND AND SHARE COMPARED

Feature	Bond	Share
Holder is	A lender	An owner
Payment	Fixed interest	Dividend, not promised
Repayment	Face value at maturity	None; sell to another buyer
Voting rights	Usually none	Usually yes
Claim if firm fails	Paid before owners	Paid last
Risk to holder	Lower	Higher

Read each row across: it states how a bond and a share differ on one point. Because a share is riskier, its required return is usually higher.

The value of a share is again a present value. If the next dividend is D , the required return is r , and dividends grow forever at a steady rate g , then value = $D / (r - g)$. With $D = \text{Tk. } 5$, $r = 12\%$ and $g = 4\%$, value = $5 / (0.12 - 0.04) = \text{Tk. } 62.50$. If the dividend does not grow, value = D / r . The formula works only when r is greater than g , and it is only as good as the forecasts that go into it.

CLOSING THE CHAPTER**SUMMARY**

A security is worth the present value of the cash it is expected to pay, at the return investors require. A bond is priced from its coupon, face value, maturity and required return. Its status depends on coupon against required return. A share has no fixed payment and is valued from its expected dividends.

THINK IT THROUGH

A bond has face value Tk. 1,000, coupon 8% paid yearly, 3 years to maturity. Find its value when the required return is 10%, and state its status.

WHERE PEOPLE GO WRONG

Students use the total of payments as the price, or use the annual rate for half-yearly interest. Others think a rise in market rates raises bond prices.

A share is a piece of ownership, with no fixed repayment.

Share. A unit of ownership in a company; also called common stock.